Graph Neural Networks to Detect Fake News

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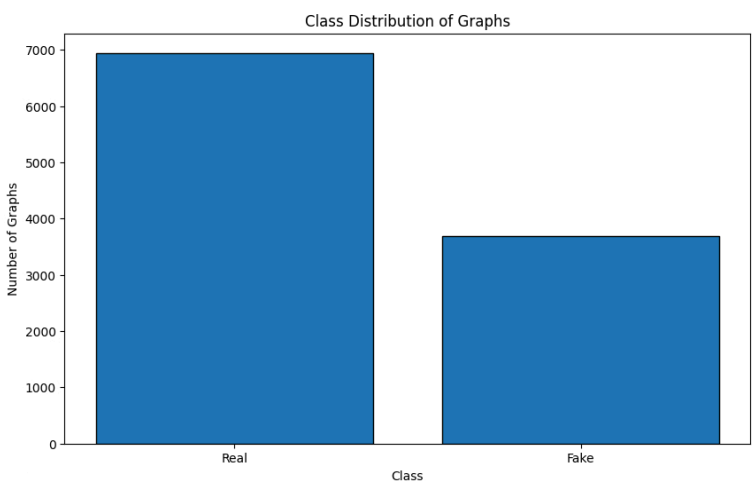
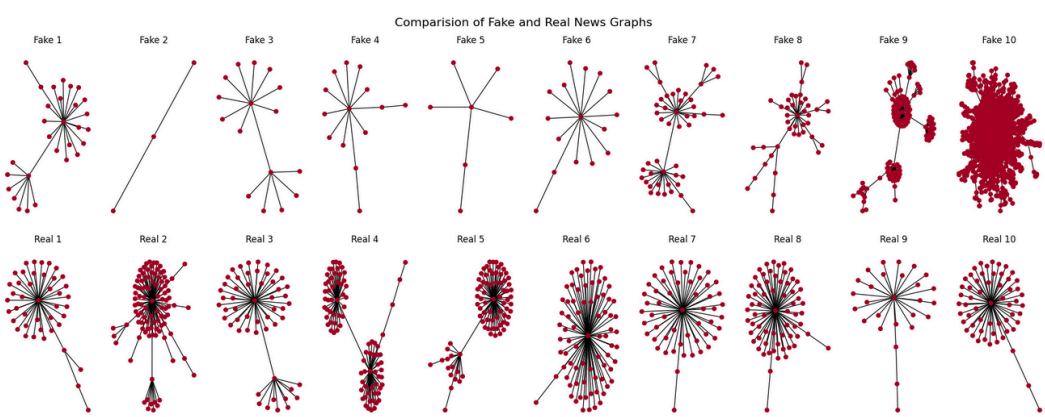
[Sharon-Bures/Deep-Learning-Project]

Data Presentation and Preliminary Analysis

Acquiring data was a challenge. The data must be a graph of social interactions about news and it also must be labeled fake and real. An attempt was made to collect data from Reddit using PRAW, Reddit's API, from 2 subreddits, those being r/worldnews and r/conspiracy. All posts from r/worldnews were labeled real and r/conspiracy fake. However, this was unsuccessful as the networks were not different enough between the 2 subreddits, therefore the model did not learn any patterns but simply labeled all networks as either fake or real. Eventually the full FakeNewsNet dataset from GossipCop with the entire networks as opposed to only the tweet ids and labels was found, and this was used instead. This dataset is labeled fake and real news from tweets and contains the network of retweets of the original tweet. The idea is to examine the structure of the network to find differences in the way fake and real news propagates on Twitter. Due to it being a professionally collected and curated dataset, there were no missing values, errors, or formats that needed to be adjusted. Only the graphs needed to be converted into a format that Pytorch Geometric could take. This dataset has no metadata, just the user interaction of retweets, therefore there is no text data that needs to be normalized or categorical data to be dealt with. In total there are 10,629 graphs, of which 6,945 are real news and 3684 are fake, therefore about twice as many real news as there are fake news graphs. Below are examples of fake and real news graphs.





Deep Learning Model Implementation

Model Architecture

Two types of graph neural networks were tested, those being a graph convolutional network (GCN) and graph attention network (GAT). Both are specifically designed for node classification tasks by utilizing the graph structure and node features. After experimenting with several architectures, the following designs achieved the highest validation accuracy. The GCN model consists of two graph convolutional layers, with an intermediate dropout layer to mitigate overfitting. The model uses an input dimension of 2, which includes node features and an additional feature representing the level of retweet. The first graph convolutional layer transforms these input features into a hidden dimension of 16, followed by the ReLU activation function to introduce non-linearity. To prevent overfitting, a dropout layer with a dropout rate of 0.5 is applied to the output of the first convolutional layer. The second convolutional layer then maps the hidden features to the output dimension, which corresponds to the two classes (fake or real) in this classification task. The output is processed through a log-softmax function to produce class probabilities for each node. The GAT model enhances the GCN architecture by incorporating attention mechanisms, allowing the model to learn the relative importance of neighboring nodes within the graph. Like the GCN, the GAT model uses an input dimension of 2 and a hidden dimension of 16. The first layer is a graph attention layer that computes attention coefficients between nodes and aggregates features accordingly. This layer is parameterized by 4 attention heads, which transform the input features into hidden features, followed by the ELU activation function for non-linearity. The second graph attention layer further processes the features and reduces them to the output dimension of 2. The final output is passed through a log-softmax function to generate class probabilities.

Training Process

In order to ensure that the model has a balanced representation of both real and fake news and to minimize bias, we first balanced the dataset by randomly oversampling the minority class (fake news) to match the size of the majority class (real news). This approach provided an equal number of fake and real news samples and prevented the model from becoming biased towards the majority class.

The balanced dataset was then shuffled and split into three subsets: training, validation, and testing. For the training subset we used 70% of the data, so the model had sufficient instances to learn the patterns and features necessary to differentiate between real and fake news. The validation set (15%) was employed to tune the hyperparameters and monitor the model's performance during training. It was essential for evaluating how well the model generalizes to unseen data and for early stopping, which helps in preventing overfitting. The testing set (15%) was reserved for the final evaluation of the model after training. This set provided an unbiased evaluation of the model's performance.

To furthermore counteract overfitting, we employed an early stopping mechanism. This approach involved monitoring the validation loss during the training process. If the validation loss did not improve for a patience of 10 epochs, the training was halted, and the model parameters from the epoch with the lowest validation loss were retained. This strategy ensured

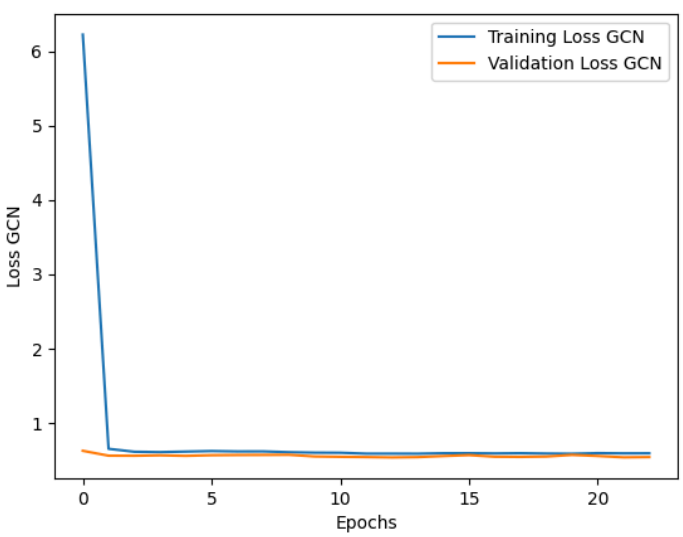
that the model did not continue to learn noise or irrelevant features, thereby improving its ability to generalize.

The effectiveness of our GCN model in distinguishing between real and fake news articles was evaluated through extensive hyperparameter tuning. We experimented with different configurations of learning rates, weight decay, and minimum delta values to identify the optimal settings that would balance model accuracy with generalizability. Specifically, we tested a learning rate (LR) of 0.0001, 0.001 and 0.01, which controlled the step size during the gradient descent optimization process. To regulate model complexity and mitigate overfitting, we employed weight decay values of 0.00001, 0.0001, and 0.001. Furthermore, we examined minimum delta values of 0.00001, 0.0001, and 0.001 for our early stopping mechanism. By systematically varying these hyperparameters, we assessed their impact on training and validation loss, and validation and test accuracy. The results of these experiments were documented, with accuracy metrics calculated for both the validation and test datasets to provide a comprehensive view of the model's performance across different configurations. This approach allowed us to fine-tune the model parameters effectively, ensuring that the final model configuration achieved high accuracy while maintaining its ability to generalize well to unseen data. We plotted the training and validation losses across epochs to visualize the training.

Research Analysis

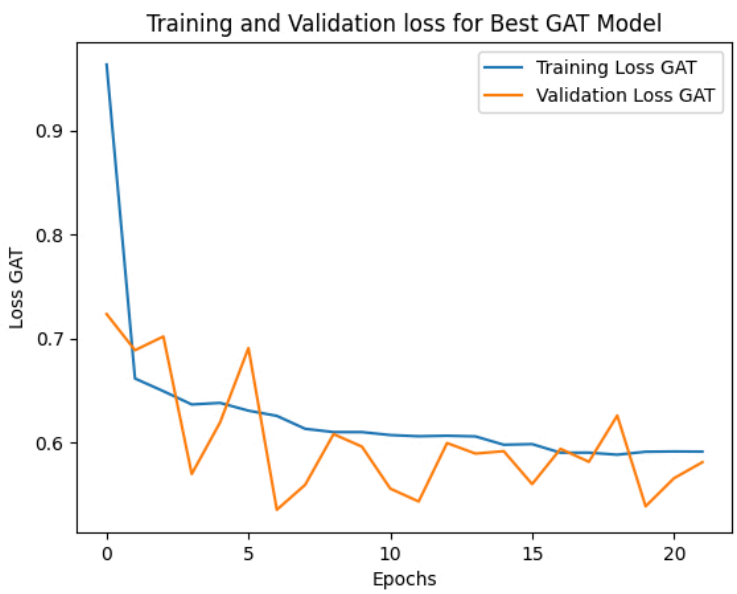
Results

The best performing GCN model was the one where the learning rate was set at 0.001 and both the weight decay and the minimum delta were set at 0.0001. The model achieved an accuracy of 83.6% on the validation set and 82.8% on the test set. The loss curve of the model shown in the Figure reveals that the training loss started high but rapidly decreased and stabilized. The Validation loss was low and stable throughout the entire training period, this indicates that the model effectively learned from the data without significant overfitting. The use of early stopping ensured that the training halted after 23 epochs when the validation loss ceased to improve, thereby preventing overfitting and promoting generalization. The precision, recall and F1-score for class 0 (real news) were 0.6959, 0.7137 and 0.7047, while for class 1 (fake news) these metrics were higher at 0.8905, 0.8819 and 0.8862. The weighted average F1-score was 0.8363, indicating a solid overall performance, with the model showing a slight bias towards correctly classifying fake news. The confusion matrix revealed that the model misclassified 33578 instances of fake news as real news and 36570 instances of real news as fake news. On the test set the precision, recall and F1-score for the



fake news were lower than on the validation set at 0.6815, 0,6593, and 0.6702. For the real news the metrics remained robust with a precision of 0.8781, recall of 0.8884, and F1-score of 0.8832. The weighted average F1-score on the test set was 0.8266, showing a minor drop in performance compared to the validation set. Also here the model shows a slight bias towards correctly classifying fake news. The confusion matrix for the test set indicated that the model misclassified 41313 instances of real news as fake news and 37366 instances of fake news as real news. Overall the GCN model demonstrated strong performance, but it performed better in identifying fake news than real news, as reflected by the higher precision, recall and F1-scores for the fake news class.

The GAT model with the best performance was when both the learning rate and the minimum delta were set at 0.001. The validation accuracy was 81.3% and the test accuracy was 79.9%. This model performed better on the fake class than the real class in all the metrics. For the fake class on the test set, the precision was 0.8705, recall was 0.8138, and F1 was 0.8412, while for the real class the precision was 0.5379, recall was 0.6230, and F1 was 0.5656. The closeness of the values for precision and recall within a class show that the model is not only effective at correctly identifying fake news, i.e. high precision, but also has a high rate of true positives with high recall. This is highly beneficial and a sign of a good model; a model with high precision and low recall would be excellent at labeling the graphs as fake, but there would be many true positive values that would be overlooked, meaning that out of the graphs that are labeled fake most would be actually fake but many models that are actually fake would be missed and labeled as real. Conversely, a model with high recall and low precision would label many graphs as fake and find most of them but with many false positive fake graphs, or graphs that are real and labeled as fake. Between classes however the model suffers. Most of the posts labeled fake are correct, but several of the fake graphs are labeled real seen in the low recall for real news, and the precision for real news also suggests real posts are being labeled as fake which is not ideal. As in the previous GCN model, early stopping was triggered, here after 22 epochs, therefore preventing overfitting of the model. The validation loss was not as smooth here but is close to the training loss showing that the model has not overfit on the validation data.



Conclusion

The GCN model outperformed the GAT model, especially in accurately classifying real news. It achieved higher validation and test accuracies compared to the GAT, with better overall precision, recall, and F1-scores. This suggests that the GCN model is more effective at handling

both types of news, reducing misclassification rates. Therefore, we choose the GCN model over the GAT model as it is more dependable in choosing the correct class for each graph.

While the results of this study demonstrate that graph neural networks, such as GCN and GAT, can effectively differentiate between real and fake news based on the propagation patterns in social media networks, there are several caveats to consider. One significant limitation is the reliance on social network data, which inherently depends on user interactions and engagement patterns. On a platform like X the network graph is very indicative of the likelihood of a post being fake or real, however the issue is the graph can only be seen after a post has been shared, by which point many people will have seen it and spread it more. Perhaps looking into a poster's past could help. If a user has many posts that seem to be fake in the past then it is likely they are still posting many fake news posts. On other platforms such as clickbait news sites, the graph network analysis is much harder to follow. People may share the link, but once this is off the site it's hard to track. Alternative approaches are needed, such as sentiment analysis and textual fake news detection based on the actual article. X also now does not allow easy access to the API, so this makes it difficult to apply the findings of this study in a practical, real-world setting where new posts need to be evaluated in real-time to prevent the spread of misinformation. It suggests a need for cooperation with social media platforms to enable responsible research access to data for the purpose of public safety and misinformation control.

Societal Impact

Fake news is becoming easier to create with generative AI available for everyone to create text and images that are fairly convincing. Many people are either not media literate, meaning they do not think critically about whether a text or an image is fake, or they simply accept it as it as it agrees with their viewpoints and biases. Social media is helping to create echo chambers where people only see posts that agree with them, and with fake news added in this can become very dangerous. Already we have seen elections skewed based on fake news and policies debated with fake news as a huge proponent for or against an argument such as anti-vax fake news. Fake news also leads to more extreme viewpoints, creating greater divides and makes it harder to achieve compromises in between the extremes. If a platform like X, formerly known as Twitter where this data came from, had labels on tweets suggesting that there was fake news, this could help slow the spread of fake news. X did at one point have this feature which led to many tweets from people in power being labeled as fake or misleading.

The last consideration with regards to any form of deep learning is the environmental impact. In our analysis we used very minimal GPU power as the dataset is not extremely large, however if one were to train and run neural networks on a much larger dataset the GPU power could be quite high. This is a lot of electricity being used, and if the electricity is not from green energy sources then deep learning models contribute to CO_2 emissions and climate change. This is particularly important to assess when the GPUs being used are not local to the user but are somewhere in the world and therefore affecting people who are not using GPUs. Providers of GPU access should keep in mind the impact they can have on the environment.